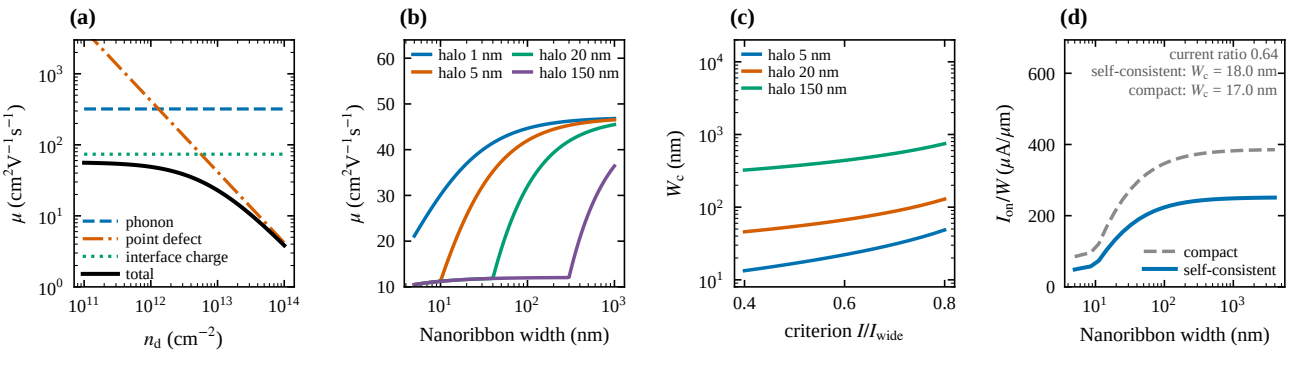
(a)
μ (cm^{2}V^{-1}s^{-1})
n_d (cm^{-2})
phonon
point defect
interface charge
total
(b)
halo 1 nm
halo 5 nm
halo 20 nm
halo 150 nm
μ (cm^{2}V^{-1}s^{-1})
Nanoribbon width (nm)
(c)
halo 5 nm
halo 20 nm
halo 150 nm
W_c (nm)
criterion I/I_{wide}
(d)
current ratio 0.64
self-consistent: W_c = 18.0 nm
compact: W_c = 17.0 nm
I_{on}/W (μA/μm)
compact
self-consistent
Nanoribbon width (nm)